DEVELOPER PORTFOLIO · 2026

Gabriel Coelho de Araújo

Software Developer | Frontend & Fullstack

RECIFE, BRAZIL · PT / EN / ES

I build working software end to end: React interfaces designed around real tasks, Flask REST APIs with documented endpoints, and Python pipelines that pull public data into relational databases.

My projects range from a solo shopping-list app shipped as an Android build to the manager module of a six-person Kotlin capstone, where I also coordinated the team. Years of tutoring taught me to explain technical decisions clearly and to work with method. I complete my degree in Systems Analysis and Development in December 2026.

LINKS

https://biieru.github.io

https://github.com/Biieru

https://linkedin.com/in/gabrielcoelho-dev

https://dev.to/bordercansado

coleho.biel@gmail.com

Scan to open the online portfolio

FRONTEND

`React` `JavaScript` `Vite` `HTML5` `CSS3`

MOBILE

`Kotlin` `Android` `Capacitor`

BACKEND & DATA

`Python` `Flask` `REST` `SQLAlchemy` `pandas`

DB & TESTING

`SQLite` `MySQL` `pytest` `unittest`

SELECTED WORK

01	**Feira Plus+**	Shopping list that sorts items by store aisle · React, Capacitor	p. 2
02	**BooksAPI**	Layered REST API with Swagger docs and pytest · Flask, SQLAlchemy	p. 3
03	**etlBCB**	ETL and forecasting on Central Bank of Brazil open data · Python	p. 3
04	**Canes**	Strategic management Android app, team of six · Kotlin, SQLite	p. 4

01 · SOLO PROJECT

Feira Plus+

A shopping list that rearranges itself to follow the aisles of the store you are in, so a trip to a wholesale market becomes a single walk through the building.

ROLE
Design and development

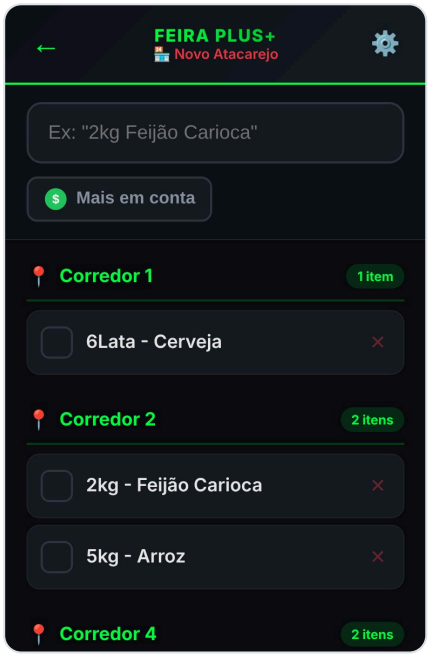
PLATFORM
Mobile web app with Android build

STACK
`React` `JavaScript` `Vite` `Capacitor` `Client-side storage`

LINKS
Source code ↗
Android APK ↗



Store selection. Each of the three supported chains has its own aisle layout.



Aisle-sorted list. Typed entries like “2kg feijão carioca” are parsed and placed in their aisle.



Rule editor. Users edit aisle names and keywords to match the real store.

THE PROBLEM

Ordinary list apps keep items in the order they were typed. In large wholesale stores (Novo Atacarejo, Assaí, Atacadão) that means crossing the floor several times. The stores offer no public product API, so the app cannot depend on their systems.

WHAT I BUILT

- Free-text item entry with quantity and unit parsing
- Automatic grouping by aisle for the selected store
- Check-off, a “better deal” flag and autocomplete from past items
- Export to plain text and to an image
- Dark, high-contrast UI with large type for one-handed use in the store

TECHNICAL DECISIONS

- **Rules instead of integration.** Each store maps to an ordered list of aisles with keywords. An item takes the first matching aisle; unmatched items go to an “Other” group that always sorts last.
- **Tolerant input parsing.** Three regular-expression patterns accept “2kg rice”, “2 rice” and “rice 2kg”. A lookup table normalizes unit spellings (kilo, quilos → kg; garrafa → bottle) and defaults to one unit.
- **Editable configuration.** Aisle rules and item history persist locally, so users adapt the layout to their store without an app update.
- **No export library.** The image export renders the list through an SVG `foreignObject` instead of adding a screenshot dependency.

Screenshots captured from the published component running in a 390 px mobile viewport. The repository currently contains the main React component and the Android build; the full Vite/Capacitor project scaffold is not yet published.

02 · SOLO PROJECT

BooksAPI

ROLE	TYPE	STACK	LINKS
Full development	REST API	`Python` `Flask` `SQLAlchemy` `SQLite` `Flasgger` `pytest`	Source code ↗

A book catalogue API with full CRUD, structured so the code can grow beyond a single file.

TECHNICAL DECISIONS

- **Application factory.** `create_app(config)` lets production and tests use different databases without code changes.
- **Layered code.** Blueprint routes, controllers and SQLAlchemy 2.0 typed models in separate modules.
- **Docs next to code.** Endpoint specs live in docstrings and are served as Swagger UI at `/apidocs`.
- **Isolated tests.** pytest fixtures run the app on in-memory SQLite; coverage tracked with pytest-cov (72% of lines).



Swagger UI generated from the code. Five endpoints: list, create, read, update and delete. Captured from a local run of the repository.

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03 · SOLO PROJECT

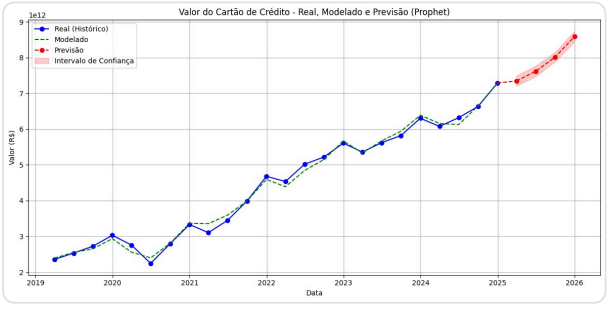
etlBCB

ROLE	TYPE	STACK	LINKS
Full development	ETL pipeline and analysis	`Python` `requests` `pandas` `SQLAlchemy` `SQLite / MySQL` `scikit-learn` `Prophet`	Source code ↗

Collects quarterly payment-method statistics (Pix, cards, transfers, boletos) from the Central Bank of Brazil open data service and turns them into a queryable dataset.

HOW IT WORKS

- **Extract.** Queries the BCB OData endpoint by quarter.
- **Transform.** Flattens the JSON with pandas and parses quarter dates.
- **Load.** Interchangeable loaders write the same DataFrame to CSV, SQLite or MySQL (SQLAlchemy), keeping storage separate from extraction.
- **Analysis.** Notebooks compare Pix and card usage, select predictors by correlation, rank features with Random Forest and forecast credit card volume with Prophet.



Notebook output. Quarterly credit card volume: historical values, Prophet fit and four-quarter forecast with confidence interval (MachineLearning.ipynb).

04 · CAPSTONE PROJECT · TEAM OF SIX

Canes

An Android app for tracking an organization's strategic actions: support staff register activities, coordinators approve them, and managers follow progress and export reports.

ROLE	CONTEXT	STACK	LINKS
Project manager and developer	Senac PE capstone, 2025	`Kotlin` `Android SDK` `SQLite` `MPAndroidChart` `iText 7`	Source code ↗ Android APK ↗

MY CONTRIBUTION

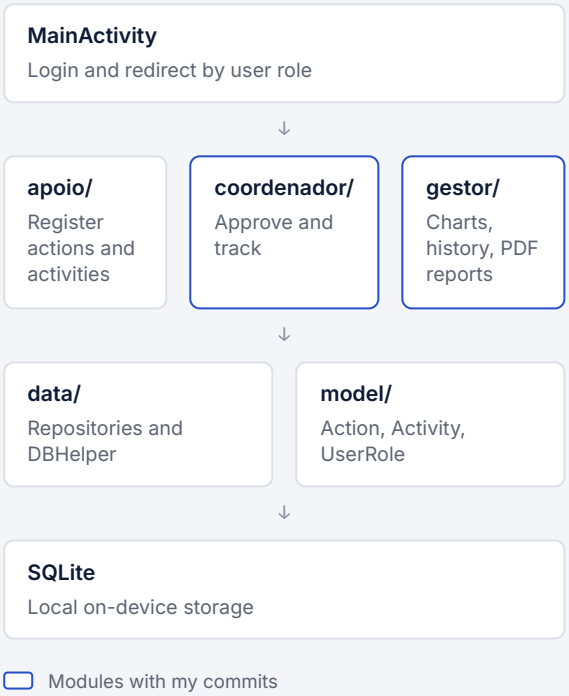
Project management. Led planning and task coordination for the six-person team through to delivery.

Development. Built most of the manager (Gestor) module and worked on the coordinator approval flow:

- Manager dashboard with pie charts of action status per strategic pillar
- PDF report generation with iText, shared through Android's `FileProvider`
- History screen with three combined filters: record type, status and owner role
- Detail screens for actions and activities
- Status filtering and repository queries for the coordinator's list of activities awaiting approval

TECHNICAL NOTES

The app routes each user at login to a role-specific area (Support, Coordinator, Manager), each in its own package. Data access goes through repository classes over a shared SQLite helper, which let several people build screens in parallel against the same data layer.



Architecture drawn from the repository's package structure. App screenshots are not included here; the APK is available in the repository releases.

ALSO BUILT

Portfolio website

Bilingual (PT/EN) single-page site with keyboard-navigable menus, visible focus states, a skip link and reduced-motion support. Deployed to GitHub Pages by a GitHub Actions workflow on every push.

`React 19` `Vite` `GitHub Actions`

https://biieru.github.io

More data pipelines

Further ETL projects on Central Bank data, including exchange rates from the PTAX API and card transactions, organized into services, preprocessors and repositories and stored in MongoDB.

`Python` `PyMongo` `MongoDB`

https://github.com/Biieru

Caminho das Constelações

An original tabletop RPG system with d100 rules and narrative points, presented on its own showcase website.

`Game design` `Web`

Showcase site ↗

PROFILE

Skills and experience

TECHNICAL SKILLS

FRONTEND

React · JavaScript · Vite · HTML5 · CSS3 · Accessible UI

MOBILE

Kotlin · Android SDK · Capacitor

BACKEND

Python · Flask · REST APIs · SQLAlchemy · Swagger / OpenAPI

DATA

SQL · SQLite · MySQL · pandas · Jupyter · scikit-learn · Prophet

TESTING

pytest · unittest · pytest-cov

TOOLS

Git · GitHub · GitHub Actions · Figma · Android Studio · VS Code

EXPERIENCE

Project Manager and Developer, Canes

2025

Senac PE capstone · team of six

Coordinated planning and task distribution through delivery while building the manager module of the Android app.

IT Teaching Assistant

During degree

Faculdade Senac Pernambuco · Recife

Answered programming questions and guided hands-on lab activities for students in IT courses.

Private Tutor

2023–2025 · 2016–2018

Self-employed · Recife

Taught elementary and middle school subjects, plus physics, history, English and writing for high school students, with lesson plans adapted to each student's pace.

EDUCATION

Associate Degree, Systems Analysis and Development

2024–Dec 2026

Faculdade Senac Pernambuco · Recife · expected December 2026

Biomedicine

2021–2023

Unibra · Recife

WRITING

Seven technical articles on dev.to (2026) covering AI for learning, gamification, augmented reality in educational games and AI governance in cybersecurity.

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ONLINE

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https://dev.to/bordercansado

LANGUAGES

Portuguese	Native
English	Fluent
Spanish	Advanced
Japanese	Basic

LOOKING FOR

Software developer roles in frontend, fullstack or web development.

Online portfolio
https://biieru.github.io